

Introduction to Probability and Applied Statistics

GOVT 6019 · PUBPOL 6019 · Cornell University · Fall 2026

Professor Melissa Sands

Course information

Course numbers	GOVT 6019 · PUBPOL 6019
Title	Introduction to Probability and Applied Statistics
Credits	4 credits
Prerequisites	None
Term	Fall 2026
Seminar	Fridays, 2:00–4:30pm
Discussion	Wednesdays, 12:20–1:10pm
Instructor	Melissa Sands · m.sands@cornell.edu
Office	201 White Hall
Instructor office hours	Wednesdays, 1:30-2:30pm, or by appointment
Course website	https://govt6019.github.io/website/
Teaching Assistant	Izza Malik · im423@cornell.edu
TA office hours	Thursdays, 10:00-11:00am, B13 White Hall

Registration note. Students enroll in one seminar section and one discussion section. Both meetings are required.

Course description

This course is the first in the Government department’s quantitative methods sequence, and serves students in Government and in the Brooks School alike. It takes you from the basics of data and measurement, through probability and statistical inference, to the linear regression model and its most useful extensions. By December you will be able to read the empirical core of most published quantitative work in political science: you will know what a regression table says, what had to be true for it to say that, and how to estimate, interrogate, and present one yourself.

The course balances two goals. First, **foundations**: we derive the central results — the behavior of sample means, the properties of estimators, the OLS estimator and its variance — because the derivations are what make later methods courses (and research) legible. Second, **practice**: the seminar is paired with hands-on work in R, built around cleaning, wrangling, and analyzing real political data.

There are **no prerequisites** for this course. No statistics background is assumed. A **math camp** for PhD students in Government, covering the algebra, calculus, and matrix notation we rely on

runs before term begins; the course introduces matrix tools as we need them.

Learning outcomes

By the end of the course, students will be able to:

1. Summarize, visualize, and describe political data, and recognize what a given measure can and cannot support.
2. Clean, manage, and analyze real datasets in R, and reproduce the analyses presented in the course.
3. Use probability and random variables to represent a data-generating process.
4. Explain why estimators have sampling distributions, and assess an estimator's bias, variance, and consistency.
5. Construct and interpret standard errors, confidence intervals, and hypothesis tests, and state precisely what they mean.
6. Derive, estimate, and interpret the linear regression model, and state the assumptions its interpretation requires.
7. Read the empirical core of published quantitative research critically, and present their own results clearly in prose, tables, and figures.

Structure

- **Seminar** (Fridays, 2:00–4:30pm)
- **Discussion section** (Wednesdays, 12:20–1:10pm, led by Izza Malik) Bring a laptop with R and Positron installed.
- **Homework** (weekly, *ungraded*): practice problems mixing short analytical exercises with applied data analysis. Solutions are released the following week. The exams look like the homework — skipping it is a strategy with a well-known sampling distribution and a bad expected value.

How the two meetings fit together. Discussion meets on Wednesday, seminar on Friday, so discussion typically (though not always) runs **the previous seminar's** material. Read the discussion dates in the calendar below accordingly.

Discussion is an integral part of the course, not a supplement you can skip: it is where most of the computing happens, and where the material you got stuck on is best untangled.

Assessment

Component	Weight	When
Midterm exam (in class)	50%	Friday, October 16 , in the seminar slot
Final exam (in person)	50%	Exam period, December 11–19 (date set by the Registrar)

The exams follow the unit structure of the calendar: the midterm covers **Units 1–3** (Weeks 1–6, data through inference), and the final covers **Unit 4** (Weeks 8–12, linear regression).

Your final grade will be determined according to the following system:

Percentage	Grade	Percentage	Grade
94% and above	A	74–76%	C
90–93%	A–	70–73%	C–
87–89%	B+	67–69%	D+
84–86%	B	64–66%	D
80–83%	B–	60–63%	D–
77–79%	C+	0–59%	F

Software

We use **R** with Positron, Posit’s data science IDE. An early discussion section walks through installation and setup. All course code is provided and all figures in the seminar are made in R — by the end of the semester nothing on a slide will be something you can’t reproduce.

Readings

Everything you strictly need is in the seminar slides, discussion materials, and homework. The course is nonetheless built around two texts, with three free companions carrying the R and presentation work. The calendar below maps readings to each week: Gailmard and A&M run throughout, and the companions appear where they are most useful.

Primary text

- **G** — Gailmard, *Statistical Modeling and Inference for Social Science* (Cambridge UP, 2014). The spine: measurement, data-generating processes, probability, sampling distributions, testing, and estimation, in close to the order we take them.

Technical backstop

- **A&M** — Aronow & Miller, *Foundations of Agnostic Statistics* (Cambridge UP, 2019). Where our derivations come from, done carefully and without parametric assumptions. Reach for it when you want the general statement behind a result.

Free companions — all three are readable online at no cost:

- **R4DS** — Wickham, Çetinkaya-Rundel & Golemund, *R for Data Science* (2nd ed.), <https://r4ds.hadley.nz/>. The reference for the data-wrangling half of the course: transforming, tidying, factors, missing values, and graphics.
- **MtM** — Arel-Bundock, *Model to Meaning* (CRC Press, 2026), <https://marginaleffects.com/>. How to turn a fitted model into a quantity you can actually interpret — predictions, comparisons, and slopes. Parts I–II only; the later case studies go well beyond this course.
- **Healy** — Healy, *Data Visualization: A Practical Introduction*, <https://socviz.co/>. Figures, and how to present model results honestly.

Calendar

All material and dates are subject to change with prior notice.

The course runs in four units:

Unit	Weeks	Dates
1 · Data	1–2	Aug 28 – Sep 11
2 · Probability	3–4	Sep 18 – Sep 25
3 · Estimation and inference	5–6	Oct 2 – Oct 9
<i>Midterm exam</i>	7	Oct 16
4 · Linear regression	8–12	Oct 23 – Dec 4

Wk	Seminar (Fri)	Topic	Discussion (Wed)	Reading
1	Aug 28	Data and measurement	Sep 2	G 1, 2.1; R4DS <i>Whole game</i>
—	<i>Sep 4</i>	<i>No seminar — APSA</i>	Sep 9 —	—
			meets	
2	Sep 11	Describing empirical data	Sep 16	G 2.2–2.3; R4DS <i>Exploratory data analysis, Factors, Missing values</i>
3	Sep 18	Probability and data-generating processes	Sep 23	G 3–4; A&M 1.1–1.2
4	Sep 25	Expectation and the sample mean	Sep 30	G 5–6; A&M 1.3, 2
5	Oct 2	Estimation and large samples	Oct 7	G 7, 9.2; A&M 3
6	Oct 9	Inference	Oct 14	G 8, 9.1; A&M 3
7	Oct 16	Midterm exam	Oct 21	—
8	Oct 23	Linear regression I	Oct 28	G 2.3.5–2.3.6; A&M 4.1
9	Oct 30	Linear regression II	Nov 4	G 7.5; A&M 4.1–4.2
10	Nov 6	Linear regression III	Nov 11	G 2.3.7, 6.7; A&M 4.4; MtM <i>Slopes</i>
—	<i>Nov 13</i>	<i>No seminar</i>	Nov 18 —	—
			meets	
11	Nov 20	Inference for regression	Dec 2	G 8.5; A&M 4.2–4.3
—	<i>Nov 27</i>	<i>No seminar — Thanksgiving Break</i>	<i>Nov 25 — none</i>	—
12	Dec 4	Presenting and interpreting results	—	MtM I–II; Healy <i>Work with Models</i> ; A&M 4.5, 7
—	Dec 11–19	Final exam (exam period)	—	—

Calendar notes.

- **Fri Aug 28** — first meeting of the course. There is no discussion section in the week of Aug 24.
- **Wed Sep 2** — first discussion section.
- **Fri Sep 4** — **no seminar** (APSA). The discussion section still meets on Sep 9. One of the two early sections (Sep 2 or Sep 9) will be devoted to course setup.
- **Sat Oct 10 – Tue Oct 13** — Fall Break.

- **Fri Oct 16** — Midterm in class.
- **Fri Nov 13** — **no seminar**. The discussion section still meets on Nov 18, taking a second sitting on Week 10's material.
- **Wed Nov 25 – Sun Nov 29** — Thanksgiving Break. Both meetings are cancelled; Week 11's discussion moves to Dec 2.
- **Fri Dec 4** — last seminar.

Course policies

Computer requirements

This course requires that you have access to a computer that can access the internet. You will need to have access to, and be able to use, the following software:

- A web browser (Google Chrome, Mozilla Firefox, Safari, etc.)
- R (free: <https://cloud.r-project.org/>) and Positron (free: <https://positron.posit.co/download.html>)

You are responsible for having a reliable computer and internet connection throughout the course.

Subject to change

All material, assignments, and dates are subject to change with prior notice. It is your responsibility to stay in touch with the teaching staff, review the course site regularly, or communicate with other students, to adjust as needed if material or dates change.

Communicating with the teaching staff

This course uses a “three before me” policy for student-to-faculty communication. It will help you find answers before we can get back to you, and it spares the teaching staff from answering the same question many times. When questions arise, please check these three sources before emailing:

1. The syllabus and your course materials
2. Announcements on the course website
3. Your classmates

Questions about homework and course computing are usually best raised in the Wednesday discussion section, then in office hours. Learning to localize a problem before asking is itself a research skill. If you have a question of a personal nature — a personal emergency, a question about a grade, or something else that needs to be communicated privately — please email.

Collaboration and homework

Working together on homework is strongly encouraged. Talking through problems is how statistics is learned. While there may be some overlap in code as you all learn the same approaches, the work you do should be your own. Exams are strictly individual.

Homework is ungraded, so there is no late penalty, and the suggested dates are posted on the course website rather than in this document. Those dates exist for a reason, though: worked solutions go out shortly afterward, and a problem you attempt only after reading the solution teaches you very little. Keep pace with them so the material is fresh when we build on it.

Both examinations are taken in person at the scheduled time; make-ups are granted only for documented, excused absences, arranged in advance where possible.

Grade disputes

In the event that a student wishes to dispute their grade on an examination, the following procedure may be used. First, students must wait at least 48 hours after the exam has been handed back. Second, students can email me to set up an appointment. This email should include a separate, typed summary of why they believe the grade is unfair. After meeting in person to discuss the exam, I will then reevaluate the grade. Revised grades may be higher or lower than the original, and the new grade will be final.

Academic integrity

Each student in this course is expected to abide by the Cornell University Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work. Students are strictly forbidden from buying or selling course materials. Cheating and plagiarism will be treated according to Cornell University regulations, which I will strictly observe. Material drawn from research materials must be cited and, if a direct quote, demarked by quotation marks. The penalty for violating the Code of Academic Integrity is an F for the examination.

The Code of Academic Integrity is in effect throughout the semester. By taking this course, you affirm that it is a violation of the code to cheat on exams, to plagiarize, to deviate from the instructor's instructions about collaboration on work that is submitted for grades, to give false information to a faculty member, and to undertake any other form of academic misconduct.

Please review the Code of Academic Integrity and university policies on plagiarism at <https://theuniversityfaculty.cornell.edu/dean/academic-integrity/code-of-academic-integrity/>. If you have any doubts about what constitutes a violation, please ask.

Artificial intelligence (AI) policy

Current University policy provides that “policies concerning the use of generative AI (GAI) tools will be decided on an assignment-by-assignment basis,” communicated with each assignment, and that adherence to course and University academic-integrity policy is required regardless of whether GAI use is permitted. For this course, the default policies are:

- **Examinations (midterm and final):** No AI tools of any kind may be used.
- **Homework and in-class coding practice:** Because AI-assisted coding tools are part of contemporary statistical practice, you may use generative AI as a *learning and coding aid* — for example, to debug an error message or to clarify a concept a second way. Since homework is ungraded, nothing here is about policing credit: the point of the homework is to build the mathematical and computational intuition the exams test, and work you outsource to AI is practice you did not get. Students who lean on AI for solutions they cannot reproduce unaided reliably find this out during the midterm.

If you are unsure whether a particular use — or whether a particular tool counts as GAI — is permitted, ask before proceeding.

Classroom behavior

Students and faculty each have responsibility for maintaining an appropriate learning environment. Students will treat one another with respect and courtesy. Disruptive students will be asked to leave the class.

Expectations and attendance

You will get out of this course what you put into it. Regular attendance and active, prepared participation are expected in both the seminar and the discussion section. Attendance is not itself graded, but much of the learning happens through working problems together, and absences bear directly on how prepared you are for the exams — which are the whole of your grade. If an extraordinary or unavoidable circumstance will cause you to miss class, email the teaching staff in advance. (If you are sick, stay home — see *Mental and physical health* below.)

Being confused out loud is useful to everyone in the room; you will never be penalized for a wrong answer offered in good faith.

Devices in class

Your attention is your most valuable asset, and devices are designed to capture it. Being present is not just about your physical presence in the classroom; you are expected to be attentive and engaged. Using laptops and smartphones for anything unrelated to the course distracts you and those around you, and is disrespectful to the class. Phones should be silenced and put away.

This course is an exception to the usual advice about laptops, but only a partial one: we write and run R code together, so bring a laptop and expect to have it open for that work. When we are not coding, close it. Otherwise pen-and-paper note-taking is preferred — working statistics by hand is part of how the material sticks, and it is how you will be doing it on the exams. Slides and materials are posted on the course website ahead of time so that you can engage in class without a screen in front of you.

Use of recording devices

Students must request permission from the instructor before making any audio or video recording of lectures, discussions, or presentations. If permission is granted, recordings and any transcripts or copies may be used only as a replacement for personal notes and may not be shared or distributed without explicit permission. This is to create a learning environment where students can feel comfortable expressing their opinions without the possibility of their contribution being reproduced, distributed, and replayed without their consent. Permission will not be withheld where a recording is a reasonable accommodation for a student's disability. This policy applies to both seminar and discussion meetings.

Accommodations for students with disabilities (SDS)

Your access to this course is important. In compliance with Cornell University policy and equal access laws, I am available to discuss appropriate academic accommodations. Please request your accommodation letter early in the semester, or as soon as you register with Student Disability Services (SDS), so that we have adequate time to arrange approved accommodations; letters should be submitted within the first two weeks of the semester where possible. Once SDS approves your

letter, it will be emailed to you and the instructor; please follow up to discuss logistics. If you experience any access barriers in this course — printed content, graphics, online materials, or communication — reach out to the instructor or your SDS counselor right away. Contact SDS as early as possible: <https://sds.cornell.edu> · 607-254-4545.

Religious observances

Students may ask for reasonable and timely accommodations for sincerely held religious beliefs. Please review the syllabus closely to determine whether your religious observance will present any scheduling conflicts with the seminar, discussion sections, or examination dates. You must inform me of any conflicts within the first two weeks of the semester.

Inclusion and diversity

It is the instructor's intent that students from all backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity students bring to the class be viewed as a resource, strength, and benefit. The instructor aims to present materials and activities that are respectful of diversity along lines of gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, and culture. I affirm people of all gender expressions and gender identities; if you prefer to be called a different name than the one indicated on the class roster, or wish to correct me on your pronouns, please let me know. Suggestions for improving the course for you or for other students are welcome at any time.

Mental and physical health

Learning happens best when you are physically and mentally healthy. If you find yourself experiencing distress that affects your studies, please consult Cornell Health. Counselors are available 24/7 at 607-255-5155, and additional support (including Let's Talk) is described at <https://health.cornell.edu>.

Coming to class sick. It is best for you — and for everyone else in the classroom — that you do not come to class when you are sick. **You will not be penalized for staying home when you are sick**, provided that you notify the teaching staff as soon as possible. While we will always try to make reasonable accommodations for you, you are ultimately responsible for catching up on material you missed by reviewing the course materials posted on the course website (including attempting the in-class programming activities when applicable) and getting notes from a peer. Wearing a mask is always available to anyone who chooses it, and if you must attend class while sick, or if you're unsure whether you're sick, please wear a mask; we can usually provide one if you ask.

University resources

This course aims to collectively create a welcoming, supportive and tolerant environment for all students and respects the various forms of diversity that they bring to the classroom. The university is committed to full inclusion in education for all persons. Services and reasonable accommodations are available to persons with temporary and permanent disabilities, students with DACA or undocumented status, students facing mental health or other personal challenges, and students with other kinds of learning challenges. Please feel free to let me know if there are circumstances affecting your ability to participate in class.

- Office of Student Disability Services: <https://sds.cornell.edu>
- Cornell Health CAPS (Counseling & Psychological Services): <https://health.cornell.edu/services/counseling-psychiatry>
- Undocumented/DACA student support: <https://dos.cornell.edu/undocumented-daca-support/undergraduate-admissions-financialaid>
- Additional academic support is available through the Learning Strategies Center, the Knight Institute for Writing in the Disciplines, and the Office of Academic Diversity Initiatives.

I respect and uphold Cornell University policies and regulations pertaining to the observation of religious holidays; assistance available to students with disabilities and to visually and/or hearing impaired students; sexual harassment; and racial or ethnic discrimination.

Cayuga Nation acknowledgment

Cornell University is located on the traditional homelands of the Gayogohó:nq' (the Cayuga Nation). The Gayogohó:nq' are members of the Haudenosaunee Confederacy, an alliance of six sovereign nations with a historic and contemporary presence on this land. The confederacy precedes the establishment of Cornell University, New York State, and the United States of America. We acknowledge the painful history of Gayogohó:nq' dispossession, and honor the ongoing connection of Gayogohó:nq' people, past and present, to these lands and waters.

This syllabus is subject to revision. Course materials prepared by the instructor may be cited but may not be sold or redistributed without permission.